آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Higher Secondary School Certificate
Examination Syllabus

Statistics

Grades XI - XII

(Based on New National Curriculum 2022-2023)

FOR ANNUAL EXAMINATION 2027 AND ONWARDS

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Higher Secondary School Certificate Examination Syllabus

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GRADES XI-XII

This syllabus will be examined in both Annual and September Examination sessions from Annual Examinations 2027 for Grade XI and Annual Examinations 2028 for Grade XII

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Preface

Established in 2002 through the Pakistan government's ordinance, the Aga Khan University Examination Board (AKU-EB) is country's first private autonomous qualification awarding body for Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSSC). Its vision is to be a model of excellence and innovation in Education in Pakistan and the developing world.

AKU-EB achieves its vision by developing examination syllabi which inculcate conceptual thinking and higher order learning and is aligned with the National Curriculum and mapped with provincial curricula and international standards. AKU-EB revises its syllabi periodically to support the needs of students, teachers and society.

The aims of the syllabus review of SSC and HSSC are to:

- Ensure continued compatibility with the goals of the National Curriculum of Pakistan.
- Review the content for inclusion of new knowledge and deletion of obsolete knowledge.
- Review the content for clarity and relevance as per the changing needs of students, teachers and society.
- Enhance and strengthen continuation and progression of content both within and across grades IX - XII (SSC and HSSC).
- Ensure the readiness of students for higher education.

During the syllabus review, the needs of all the stakeholders were identified through a needs-assessment survey. Students and teachers of AKU-EB affiliated schools from across Pakistan participated in the survey. Thereafter, a revision panel, which consisted of examiners, teachers of affiliated and non-affiliated schools, teacher trainers and university academicians, reviewed and revised the syllabus following a planned, meticulous and standardised syllabi review process.

The development of the revised syllabus has been made possible by the creativity and relentless hard work of Curriculum and Examination Development unit and the constant support provided by all the other units of AKU-EB. We are particularly thankful to our Principal Syllabus Reviewers, Syllabus Revision Panellists and all other reviewers for their contribution. We are also thankful to all the students and teachers who took part in the needs-assessment survey and to the principals of AKU-EB affiliated schools who made this endeavour possible by facilitating and encouraging their teachers and students to be a part of the survey and the syllabus revision panel.

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With your support and collective hard work, AKU-EB has been able to take the necessary steps to ensure effective implementation of the National Curriculum of Pakistan through this syllabus. We are confident that this syllabus will continue to provide the support that is needed by students to progress to the next level of education and we wish the very best to our students and teachers in implementing this syllabus.



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FOR ANNUAL EXAMINATION 2027 AND ONWARDS

Understanding of AKU-EB Syllabi

1. The AKU-EB syllabi guide the students, teachers, parents and other stakeholders regarding the topics that will be taught and examined in each grade (IX, X, XI and XII). In each syllabus document, the content progresses from simple to complex, thereby, facilitating a gradual, conceptual learning of the content.

2. The topics of the syllabi are divided into sub-topics and **student learning outcomes** (**SLOs**). The SLOs define the depth and the breadth at which each topic or subtopic will be taught, learnt and examined. The syllabi also provide enabling SLOs where needed to scaffold student learning.

3. Each SLO starts with an achievable and assessable **command word** such as describe, relate, evaluate, etc. The purpose of the command words is to direct the attention of teachers and students to specific tasks that the students are expected to undertake in the course of their studies.

4. The SLOs are classified under the following **cognitive levels** of Bloom's Taxonomy: Remember (R), Understand (U), Apply and beyond [Apply (A), Analyse (An), Evaluate (E), Create (C)]. This is to facilitate effective planning for teaching, learning and assessment. In addition, some SLOs are identified as Formative Assessments (FA), where applicable.

5. Where applicable, **Practical Activities** section is provided to elaborate the assessment in the Practical Examination.

6. The **Examination Specification** is provided which elucidates the weightage of each topic in the examinations determined on the basis of the content as well as the relevance of the topic.

7. To implement this syllabus, students and teachers can take support from additional material provided by the board to its affiliated schools including **Learning Resource Guides**, **Pacing Guides** and **Model Papers**.

8. The AKU-EB syllabi for Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSSC) are designed to foster not only conceptual understanding but also critical thinking and problem-solving skills. These syllabi ensure students develop the cognitive, affective and psychomotor skills essential for success at the university and beyond.

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Subject Rationale of AKU-EB Statistics

Why study Statistics?

Learning Statistics at higher secondary level focuses on developing statistical and critical thinking skills, enabling students to distinguish between relevant and irrelevant data. It helps them use and interpret data efficiently and effectively, and apply this understanding to solve problems, make informed decisions, and analyse real-world situations.

What will you learn in AKU-EB Statistics?

This syllabus equips students with the tools and concepts to use statistics effectively and make inferences from data. It develops the ability to analyse information, solve problems, and interpret findings in real-life contexts. Assessment focuses on applying statistical methods rather than reproducing learned material.

Where will it take you?

The AKU-EB syllabus of statistics will provide the conceptual basis for higher studies in many subjects. For those who pursue statistics in higher studies, various career opportunities are available such as:

- Actuarial Scientist
- Data Analyst
- Data Scientist
- Social Scientist
- Biostatistician
- Financial Analyst
- Market Researcher
- Database Administrator
- Psychometrician

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How to approach the syllabus?

The AKU-EB syllabus is designed in a clear, reader-friendly format so that students and teachers can easily understand and use it for teaching, learning, and assessment. The syllabus includes the following parts:

Subject Rationale	It is an introductory document for students.
Student Learning Outcomes (SLOs)	These guide students about what must be achieved.
Exam Specification	It guides regarding what is expected in the examination.
Practical Activities	These include laboratory activities to be performed during an academic year.
Additional Resources	
Pacing Guide	It ensures smooth transition and curricular continuity of a school's academic year. It also predicts the time and pace of the syllabus implementation.
Resource Guide	It includes teaching and learning resources for students and teachers.
Model Paper	It guides regarding exam pattern, types of questions and marking scheme.
Command Word Guide	It clarifies expectations regarding the cognitive levels and skills that should be acquired by the students, and which are assessed in examinations.

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Student Learning Outcomes of AKU-EB HSSC Statistics Syllabus

Part I (Grade XI)

Topics and Sub-topics	Student Learning Outcomes		Cognitive Level[1]		
			R	U	A and beyond
1. Introduction to Statistics	Students should be able to:				
1.1 Introduction	1.1.1	define statistics;	*		
	1.1.2	describe uses of statistics and its importance in different fields;		*	
	1.1.3	distinguish between: a. descriptive and inferential statistics, b. statistics and parameter, c. population and sample;		*	
	1.1.4	explain the limitations of statistics;		*	
1.2 Data and its Collection	1.2.1	distinguish between: a. primary and secondary data, b. qualitative and quantitative data, c. discrete and continuous data;		*	
	1.2.2	explain the methods of collection of primary and secondary data;		*	
	1.2.3	describe the following terms: a. constant, b. variable, c. quantitative variable, d. discrete variable, e. continuous variable, f. qualitative variable;		*	

[1]R = Remember, U = Understand, A = Apply and beyond [Apply (A), Analyse (An), Evaluate (E), Create (C)]

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
		1.2.4	describe the term, 'questionnaire';		*	
		1.2.5	write the characteristics of a good questionnaire;	*		
		1.2.6	construct a simple questionnaire using *Google Forms* or *SurveyMonkey;*			FA[2]
1.3	Measurement Scales	1.3.1	explain nominal scale, ordinal scale, interval scale and ratio scale;		*	
		1.3.2	distinguish between: a. nominal and ordinal scale, b. interval and ratio scale;		*	
		1.3.3	classify variable based on its nature into nominal, ordinal, interval and ratio scale;		*	
1.4	Application of Statistics	1.4.1	explain the applications of statistics in real-world scenarios;		FA	
1.5	Application of Statistical Software in Data Analysis	1.5.1	solve practical data problems using tools like minitab and excel.			FA

[2] FA= Formative Assessment, not to be assessed under examination conditions

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Topics and Sub-topics		Student Learning Outcomes	Cognitive Level		
			R	U	A and beyond
2. Presentation of Statistical Data		Students should be able to:			
2.1 Frequency Distribution	2.1.1	describe tabulation and its classification (one way and two way classifications);		*	
	2.1.2	describe the importance of accurate data representation in statistical analysis;		*	
	2.1.3	describe the following terms: a. class interval, b. tally marks/ frequency, c. upper class limit/ boundary, d. lower class limit/ boundary, e. class mark, f. class width;		*	
	2.1.4	construct a frequency distribution table for discrete and continuous data;			A
	2.1.5	calculate relative frequency, less than cumulative frequency and more than cumulative frequency;			A
	2.1.6	interpret relative frequency, less than cumulative frequency and more than cumulative frequency;			An
	2.1.7	construct stem and leaf diagram for: a. one set of data, b. two sets of data (back to back);			A
	2.1.8	solve problems based on the stem and leaf for one and two sets of data;			A
2.2 Graphical Representation	2.2.1	draw simple bar chart, multiple bar chart, components (subdivided) bar chart and pie chart;			A
	2.2.2	draw histogram (for both equal and unequal class intervals), frequency polygon, frequency curve, cumulative frequency polygon, and cumulative frequency curve;			A
	2.2.3	interpret statistical graphs and diagrams related to SLOs # 2.2.1 and 2.2.2.			An

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
3. Measures of Central Tendency		Students should be able to:				
3.1	Central Tendency	3.1.1	define central tendency;	*		
		3.1.2	describe averages and their types;		*	
3.2	Arithmetic Mean	3.2.1	describe arithmetic mean and weighted arithmetic mean;		*	
		3.2.2	calculate the arithmetic mean for ungrouped and grouped data using formulas;			A
		3.2.3	calculate the weighted arithmetic mean;			A
		3.2.4	describe the following properties of arithmetic mean: a. if $X = a$ (a is constant), then $\bar{X} = a$, b. if $Y = X \pm a$, then $\bar{Y} = \bar{X} \pm a$, c. if Y = bX, then $\bar{Y} = b\ \bar{X}$, d. if $Y = \frac{X}{a}$, then $\bar{Y} = \frac{\bar{X}}{a}$ *or* $\bar{Y} = \frac{1}{a}(\bar{X})$, e. if $Y = aX \pm b$, then $\bar{Y} = a\,\bar{X} \pm b$;		*	
		3.2.5	Solve problems by applying the properties mentioned in SLO # 3.2.4;			A
		3.2.6	describe advantages and disadvantages of using arithmetic mean;		*	
3.3	Geometric Mean	3.3.1	describe geometric mean and its formula;		*	
		3.3.2	calculate the geometric mean for ungrouped and grouped data using formula and the logarithmic method;			A
		3.3.3	describe advantages and disadvantages of using geometric mean;		*	
3.4	Harmonic Mean	3.4.1	describe harmonic mean and its formula;		*	
		3.4.2	calculate the harmonic mean for ungrouped and grouped data;			A
		3.4.3	describe advantages and disadvantages of using harmonic mean;		*	

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
3.5	Relationship between Arithmetic Mean, Geometric Mean and Harmonic Mean	3.5.1	describe the relationship between arithmetic mean, geometric mean and harmonic mean;		*	
		3.5.2	verify the relationship between arithmetic mean, geometric mean and harmonic mean for a given data;			E
3.6	Median and Quartiles	3.6.1	describe median and quartiles for ungrouped and grouped data;		*	
		3.6.2	calculate median and quartiles for ungrouped and grouped data;			A
		3.6.3	describe advantages and disadvantages of median;		*	
		3.6.4	interpret median and quartiles through the cumulative frequency curve;			An
		3.6.5	draw box and whisker plot;			A
		3.6.6	solve problems based on the box and whisker plot for one and two sets of data;			A
3.7	Mode	3.7.1	define mode;	*		
		3.7.2	calculate mode for ungrouped and grouped data;			A
		3.7.3	describe advantages and disadvantages of mode;		*	
		3.7.4	estimate mode through histogram and stem and leaf diagram;			E
3.8	Empirical Relationship between Mean, Median and Mode	3.8.1	describe the empirical relationship between arithmetic mean, median and mode in symmetric and skewed distributions;		*	
		3.8.2	solve problems using the empirical relationship between arithmetic mean, median and mode;			A
		3.8.3	evaluate the appropriateness of majors of central tendency (mean, median, or mode) for a given context.			E

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
4. Measures of Dispersion		Students should be able to:				
4.1	Dispersion	4.1.1	describe dispersion of data;		*	
		4.1.2	distinguish between measures of central tendency and measures of dispersion;		*	
4.2	Range	4.2.1	define range and semi range;	*		
		4.2.2	calculate range for ungrouped and grouped data;			A
		4.2.3	describe advantages and disadvantages of range;		*	
		4.2.4	describe coefficient of range for ungrouped and grouped data;		*	
		4.2.5	calculate coefficient of range for ungrouped and grouped data;			A
4.3	Quartile Deviation	4.3.1	describe interquartile range, quartile deviation and coefficient of quartile deviation;		*	
		4.3.2	describe advantages and disadvantages of quartile deviation;		*	
		4.3.3	find the interquartile range, quartile deviation and coefficient of quartile deviation for ungrouped and grouped data;			A
4.4	Mean Deviation	4.4.1	describe mean deviation and coefficient of mean deviation from mean, median and mode;		*	
		4.4.2	describe advantages and disadvantages of mean deviation;		*	
		4.4.3	find the mean deviation and coefficient of mean deviation from mean, median and mode for ungrouped and grouped data;			A
4.5	Variance and Standard Deviation	4.5.1	describe variance, standard deviation and coefficient of variation;		*	
		4.5.2	describe advantages and disadvantages of coefficient of variation;		*	
		4.5.3	calculate variance and standard deviation for ungrouped and grouped data;			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
		4.5.4	calculate the coefficient of variation for ungrouped and grouped data;			A
		4.5.5	describe the following properties of variance and standard deviation: a. S.D.(a) = 0 and Var(a) = 0, b. S.D.(X + a) = S.D.(X) and Var(X +a) = Var(X), c. S.D.(X – a) = S.D.(X) and Var(X –a) = Var(X), d. S.D.(aX) = aS.D.(X) and Var(aX) = a^2 Var(X), e. S.D.$\left(\frac{X}{a}\right)=\left(\frac{1}{a}\right)$S.D.(X) and Var$\left(\frac{X}{a}\right)=\left(\frac{1}{a^2}\right)$Var(X);		*	
		4.5.6	solve problems using the properties as given in SLO # 4.5.5;			A
4.6	Moments	4.6.1	write the formulae for moments about the arithmetic mean and about the origin;	*		
		4.6.2	solve problems related to moments about the arithmetic mean and about the origin for ungrouped data;			A
4.7	Symmetry and Skewness of a Distribution	4.7.1	describe symmetry;		*	
		4.7.2	explain the following properties of a symmetrical distribution: a. mean = median = mode, b. third quartile – median = median – first quartile, c. all odd ordered moments about mean vanish;		*	
		4.7.3	explain the skewness of a distribution (positive or negative);		*	
		4.7.4	find the coefficient of skewness using Karl Pearson's formulae;			A
		4.7.5	interpret the coefficient of skewness of a distribution;			An
		4.7.6	solve problems related to coefficient of skewness and symmetry;			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
4.8	Kurtosis	4.8.1	describe kurtosis and its types;		*	
		4.8.2	calculate the coefficient of kurtosis to identify whether a symmetrical distribution is platykurtic, mesokurtic, or leptokurtic;			A
		4.8.3	solve problems related to kurtosis.			A
4.9	Application of Statistical Software in Data Analysis	4.9.1	solve practical data problems using tools like minitab and excel.			FA

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
5. Index Numbers		Students should be able to:				
5.1	Introduction to Index Numbers	5.1.1	describe index number and its types (price index, quantity index and aggregate index number);		*	
		5.1.2	write the uses of index numbers;	*		
		5.1.3	describe the steps involved in the construction of whole sale price index numbers;		*	
		5.1.4	write the limitations of index numbers;		*	
5.2	Simple Relatives or Simple Index Numbers	5.2.1	describe simple index numbers;		*	
		5.2.2	distinguish between simple and composite price index numbers;		*	
		5.2.3	calculate simple price index numbers using: a. fixed base method, b. chain base method;			A
5.3	Unweighted Index Numbers	5.3.1	describe the two methods of constructing unweighted index numbers: a. simple aggregate or aggregative method, b. simple average of relatives method;		*	
		5.3.2	calculate composite price index numbers using the method of simple aggregate or aggregative method;			A
5.4	Weighted Index Numbers	5.4.1	describe the following methods of constructing weighted index numbers: a. weighted aggregate method, b. weighted average of relatives method;		*	
		5.4.2	calculate weighted aggregative composite price index numbers using: a. Laspeyre's formula, b. Paasche's formula, c. Fisher's formula.			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
6. Time-Series		Students should be able to:				
6.1	Concept of Time-Series	6.1.1	describe time-series;		*	
		6.1.2	describe the following components of a time-series: a. secular trend, b. seasonal variations, c. cyclical fluctuations, d. irregular movements;		*	
6.2	Measurement of Secular Trend	6.2.1	define linear trend and trend values;	*		
		6.2.2	explain the technique of coding time variable;		*	
		6.2.3	describe the following methods of measuring secular trend: a. the freehand curve method, b. the method of semi-averages, c. the method of moving averages;		*	
		6.2.4	find the trend values of a given data using the methods mentioned in SLO # 6.2.3.			A
6.3	Application of Statistical Software in Data Analysis	6.3.1	solve practical data problems using tools like minitab and excel.			FA

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Topics and Sub-topics		Student Learning Outcomes	Cognitive Level		
			R	U	A and beyond
7. Regression and Correlation		Students should be able to:			
7.1 Simple Linear Regression	7.1.1	define regression;	*		
	7.1.2	describe independent variable (regressor) and dependent variable (regressand);		*	
	7.1.3	draw a scatter diagram;			A
	7.1.4	discuss the nature of the data using a scatter diagram;			E
	7.1.5	describe simple linear regression and its coefficient;		*	
	7.1.6	describe the method of least squares line of regression;		*	
	7.1.7	apply the method of least squares to fit a regression line of Y on X and X on Y;			A
	7.1.8	solve problems related to SLOs # 7.1.4 to 7.1.7;			A
7.2 Simple Linear Correlation	7.2.1	describe correlation;		*	
	7.2.2	explain the nature of correlation (positive correlation, negative correlation, non-linear correlation and no correlation);		*	
	7.2.3	distinguish between regression and correlation;		*	
	7.2.4	describe correlation coefficient 'r'(also called Pearson product-moment correlation coefficient);		*	
	7.2.5	calculate the coefficient of correlation using the following formulae: a. $r = \frac{\sum XY - n\overline{X}\,\overline{Y}}{\sqrt{\left[\sum X^2 - n\overline{X}^2\right]\left[\sum Y^2 - n\overline{Y}^2\right]}}$, b. $r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}$, c. $r = \sqrt{b \times d}$ (product of two regression coefficients);			A
	7.2.6	write the properties of the coefficient of correlation;	*		
	7.2.7	solve problems related to the coefficient of correlation,			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
7.3	Rank Correlation	7.3.1	describe rank correlation;		*	
		7.3.2	find the coefficient of rank correlation using Spearman's formula for rank correlation;			A
		7.3.3	calculate the rank correlation for tied ranks.			A
7.4	Application of Statistical Software in Data Analysis	7.4.1	solve practical data problems using tools like minitab and excel.			FA

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Part II (Grade XII)

Topics and Sub-topics		Student Learning Outcomes		Cognitive Level[1]		
				R	U	A and beyond
8. Counting Techniques		Students should be able to:				
8.1	Fundamental Principle of Counting Techniques	8.1.1	explain the concept of factorial (!);		*	
		8.1.2	solve problems based on factorial (n!);			A
		8.1.3	describe fundamental principle of counting;		*	
		8.1.4	construct a tree diagram using the fundamental principle of counting for: a. repeated values, b. non repeated values;			A
		8.1.5	solve problems related to fundamental principle of counting and tree diagram for: a. repeated values, b. non repeated values;			A
8.2	Permutation	8.2.1	describe permutation of *n* different objects taken *r* at a time and recognise the notation ${}^{n}P_{r}$;		*	
		8.2.2	solve problems based on permutation;			A
		8.2.3	solve problems related to permutation when: a. repetition is not allowed, b. repetition is allowed, c. elements are identical, d. arrangements are in a circle;			A
8.3	Combination	8.3.1	describe combination of *n* different objects taken *r* at a time and recognise the notation ${}^{n}C_{r}$;		*	
		8.3.2	solve problems based on combination.			A

[1] R = Remember, U = Understand, A = Apply and beyond [Apply (A), Analyse (An), Evaluate (E), Create (C)]

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
9. Probability		Students should be able to:				
9.1	Introduction to Probability	9.1.1	describe the following terms: a. random experiment, b. sample space, sample point and an event, c. simple and compound events, d. impossible and sure events, e. complementary events, f. equally likely events, g. mutually exclusive and mutually inclusive (non-exclusive) events, h. exhaustive events, i. dependent and independent events;		*	
		9.1.2	apply set theory concepts to random experiments, sample spaces, and events;			A
		9.1.3	calculate probability through the classical, relative frequency and axiomatic approaches;			A
		9.1.4	find the probability of occurrence of an event using Venn and tree diagrams;			A
9.2	Laws of Probability	9.2.1	explain the law of addition of probability: a. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where A and B are two non-mutually exclusive events, b. $P(A \cup B) = P(A) + P(B)$, where A and B are mutually exclusive events;		*	
		9.2.2	explain the law of multiplication of probability: $P(A \cap B) = P(A) \times P(B)$, where A and B are independent events;		*	

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Topics and Sub-topics	Student Learning Outcomes		Cognitive Level		
			R	U	A and beyond
	Students should be able to:				
	9.2.3	define 'conditional probability';	*		
	9.2.4	explain the law of multiplication of conditional probability: $P(A \cap B) = P(A) \times P(B \mid A)$ or $P(A \cap B) = P(B) \times P(A \mid B)$, where $P(B\|A)$ and $P(A\|B)$ are conditional probabilities and A and B are dependent events;		*	
	9.2.5	solve problems related to the SLO # 9.2.1 to 9.2.4. (**Note**: Problem involving Venn and tree diagrams are also included.)			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
10. Random Variables		Students should be able to:				
10.1	Concept of Random Variable	10.1.1	describe random, discrete random and continuous random variables;		*	
		10.1.2	distinguish between discrete and continuous random variables;		*	
10.2	Discrete Random Variable	10.2.1	describe the probability distribution of a discrete random variable;		*	
		10.2.2	draw a probability distribution table related to a given situation involving a discrete random variable;			A
		10.2.3	describe the probability distribution function of a discrete random variable;		*	
		10.2.4	solve problems related to probability distribution function of a discrete random variable;			A
		10.2.5	describe the expected value of a discrete random variable;		*	
		10.2.6	find the expected value of a discrete random variable;			A
		10.2.7	describe the properties of the expected value of a discrete random variable;		*	
		10.2.8	apply the properties of expected value of a discrete random variable to a given set of data;			A
		10.2.9	describe the variance and standard deviation of a discrete random variable;		*	
		10.2.10	find the variance and standard deviation of a discrete random variable;			A
		10.2.11	describe the properties of variance and standard deviation of a discrete random variable;		*	

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
10.3	Continuous Random Variable	10.3.1	describe the probability distribution and probability density function (p.d.f.) of a continuous random variable;		*	
		10.3.2	solve problems related to probability density function (p.d.f.) of a continuous random variable;			A
		10.3.3	describe the expected value and variance of a continuous random variable;		*	
		10.3.4	solve problems related to expectation, variance and standard deviation of a continuous random variables;			A
10.4	Combinations of Random Variables	10.4.1	describe the following properties about the expected value and variance for the sum and difference of two independent random variables X and Y: a. $E(X \pm Y) = E(X) \pm E(Y)$, b. $E(aX \pm bY) = aE(X) \pm bE(Y)$, c. $V(X \pm Y) = V(X) + V(Y)$, d. $V(aX \pm bY) = a^2V(X) + b^2V(Y)$;		*	
		10.4.2	solve problems related to the properties mentioned in SLO # 10.4.1.			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
11. Probability Distributions		Students should be able to:				
11.1	Uniform Distribution for discrete random variable	11.1.1	describe the discrete uniform probability distribution;		*	
		11.1.2	solve problems related to the discrete uniform probability distribution;			A
		11.1.3	calculate the mean, variance and standard deviation of discrete uniform probability distribution;			A
11.2	Binomial Distribution	11.2.1	define the following terms: a. bernoulli trial, b. binomial experiment, c. binomial random variable;	*		
		11.2.2	write the properties of binomial experiment;	*		
		11.2.3	describe the binomial probability distribution (Bernoulli distribution);		*	
		11.2.4	describe the binomial frequency distribution;		*	
		11.2.5	solve problems related to the binomial probability distribution;			A
		11.2.6	calculate the mean, variance and standard deviation of binomial probability distribution;			A
11.3	Hyper Geometric Distribution	11.3.1	define the terms: a. hyper geometric experiment, b. hyper geometric random variable;	*		
		11.3.2	write the properties of hyper geometric experiment;	*		
		11.3.3	describe the hyper geometric probability distribution;		*	
		11.3.4	solve problems related to the hyper geometric probability distribution;			A
		11.3.5	calculate the mean, variance and standard deviation of hyper geometric probability distribution;			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
11.4	Uniform Distribution for Continuous Random Variable	11.4.1	describe the continuous uniform probability distribution;		*	
		11.4.2	solve problems related to the continuous uniform probability distribution;			A
		11.4.3	calculate the mean, variance and standard deviation of continuous uniform probability distribution;			A
11.5	Normal Distribution	11.5.1	define normal random and standard normal random variables;	*		
		11.5.2	describe the normal and the standard normal distribution and their probability density functions;		*	
		11.5.3	describe the properties of a normal distribution with the help of a diagram;		*	
		11.5.4	find the probabilities of the normal and standard normal random variable using the standard normal distribution table;			A
		11.5.5	solve problems related to the normal distribution;			A
11.6	Poisson Distribution	11.6.1	describe poisson probability distribution;		*	
		11.6.2	solve problem related to poisson probability distribution;			A
		11.6.3	calculate the mean, variance and standard deviation of poisson probability distribution;			A
		11.6.4	describe the conditions for using the Poisson distribution as an approximation to the Binomial distribution;		*	
		11.6.5	calculate probabilities of random variable using poisson approximation;			A
		11.6.6	write the conditions for using the normal distribution with continuity correction as an approximation to the Poisson;	*		
		11.6.7	calculate probabilities of random variable using normal approximation.			A
11.7	Application of Statistical Software in Data Analysis	11.7.1	solve practical data problems using tools like minitab and excel.			FA

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
12. Sampling and Sampling Distributions		Students should be able to:				
12.1	Concept of Sampling	12.1.1	define sampling, sampling units, sampling frame and sampling design;	*		
		12.1.2	distinguish between: a. finite and infinite population, b. sample and population, c. parameter and statistics;		*	
		12.1.3	describe advantages and disadvantages of sampling;		*	
		12.1.4	distinguish between: a. probability and non-probability sampling, b. random sampling with and without replacement;		*	
		12.1.5	explain simple random sampling, stratified random sampling, systematic random sampling, quota sampling and cluster random sampling;		*	
		12.1.6	solve problems related to SLO # 12.1.5;			A
12.2	Sampling Distribution of Sample Mean	12.2.1	define the following terms: a. sampling distribution, b. sampling errors, c. non-sampling errors of statistics;	*		
		12.2.2	distinguish between sampling errors and non-sampling errors of statistics;		*	
		12.2.3	define the sampling distribution of sample mean;	*		
		12.2.4	describe the properties of a sampling distribution of sample mean;		*	
		12.2.5	construct the sampling distribution of sample mean;			A
		12.2.6	determine the sampling distribution of the sample mean;			A
		12.2.7	verify the properties of a sampling distribution of sample mean;			E

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
12.3	Sampling Distribution of Difference Between Two Sample Means	12.3.1	describe the sample distribution of difference between two sample means and its properties;		*	
		12.3.2	find the sample distribution of difference between two sample means;			A
		12.3.3	verify the properties of a sample distribution of difference between two sample means.			E
12.4	Application of Statistical Software in Data Analysis	12.4.1	solve practical data problems using tools like minitab and excel.			FA

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
13. Estimation		Students should be able to:				
13.1	Point Estimation	13.1.1	define estimation of parameter, point estimation and point estimator;	*		
		13.1.2	distinguish between biased and unbiased estimators;		*	
		13.1.3	find the point estimates for population mean and population variance from the given random sample;			A
13.2	Interval Estimation	13.2.1	define interval estimation of a parameter and confidence interval;	*		
		13.2.2	estimate the confidence interval for the: a. mean of a normal population (known and unknown standard deviation), b. difference between means of two normal populations (known and unknown standard deviations), c. population proportion (large sample), d. difference between proportions of two populations (large samples).			E

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
14. Statistical Inference		Students should be able to:				
14.1	Introduction to Hypothesis	14.1.1	describe statistical hypothesis and hypothesis testing;		*	
		14.1.2	distinguish between the following types of hypotheses: a. null and alternative, b. simple and composite;		*	
		14.1.3	describe null and alternative hypothesis for a given situation;		*	
		14.1.4	describe the following elements of hypothesis testing: a. test statistic, b. rejection and acceptance region, c. critical value, d. one tailed test, e. two tailed test, f. Type I and Type II errors, g. level of significance, h. decision rule, i. conclusion;		*	
14.2	Hypothesis Testing	14.2.1	apply the test of hypothesis in the following cases concerning: a. the population mean when population standard deviation is known or unknown, b. the difference between two means of two different populations when populations' standard deviation are known;			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
14.3	Attribute	14.3.1	describe attribute;		*	
		14.3.2	distinguish between variable and attribute;		*	
		14.3.3	identify the categorical data of two attributes;		*	
		14.3.4	describe the independence of two attributes;		*	
		14.3.5	define the association of attributes;	*		
		14.3.6	describe positive association, negative association, complete association and complete disassociation;		*	
		14.3.7	distinguish between correlation and association;		*	
		14.3.8	describe the coefficient of association;		*	
		14.3.9	find the coefficient of association;			A
14.4	Contingency Table	14.4.1	describe a contingency table and its degree of freedom;		*	
		14.4.2	describe the criterion of independence of two attributes in a contingency table;		*	
		14.4.3	describe the chi-square (χ^2) statistic;		*	
		14.4.4	describe the test for independence;		*	
		14.4.5	apply the chi-square (χ^2) statistic to test the association between the attributes;			A
14.5	Analysis of Variance (One way and Two way)	14.5.1	describe analysis of variance;		*	
		14.5.2	write the basic assumptions of the analysis of variance;	*		
		14.5.3	define the following terms: a. total sum of squares, b. treatment sum of squares, c. error sum of squares;	*		
		14.5.4	describe the relation of total sum of squares = treatment sum of squares + error sum of squares;		*	
		14.5.5	calculate the total sum of squares, the treatment sum of squares and the error sum of squares;			A

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Topics and Sub-topics		Student Learning Outcomes		Cognitive Level		
				R	U	A and beyond
		Students should be able to:				
		14.5.6 14.5.7 14.5.8	calculate the degree of freedom for total sum of squares, the treatment sum of squares and the error sum of squares; calculate the treatment mean square and the error mean square; use a hypothesis to test the equality of means of normal populations.			A A A
14.6	Application of Statistical Software in Data Analysis	14.6.1	solve practical data problems using tools like minitab and excel.			FA

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Practical Activities of AKU-EB HSSC Statistics Syllabus

Student Learning Outcomes

Content Covered	Actions Performed	Attitude Developed
• Data Representation • Frequency Distributions • Measures of Central Tendency • Measures of Dispersion • Regression and Correlation Analysis • Probability Distributions • Sampling Distributions • Confidence Interval Estimation • Hypothesis Testing (z-test, errors) • Analysis of Variance (ANOVA)	• Construct and interpret graphical representations of data • Organise raw data into frequency tables and calculate related frequencies • Calculate and compare measures like median, quartiles, skewness, and kurtosis • Fit regression lines and interpret correlation coefficients • Create probability distribution tables and compute mean and variance • Solve problems involving normal distribution and z-scores • Compute sampling distributions for differences in sample means • Estimate confidence intervals and apply hypothesis testing procedures • Calculate sums of squares and degrees of freedom for ANOVA	• Analytical thinking in interpreting statistical data and distributions • Precision and accuracy in calculations and data representation • Critical evaluation of data through graphical and numerical methods • Confidence in applying statistical techniques to real-world problems • Careful formulation and testing of hypotheses • Appreciation of the role of variability and uncertainty in data analysis

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Topic Wise Practical Activities

Grade XI

S. No.	PRACTICAL ACTIVITY
	Presentation of Statistical Data
1.	Construct and interpret bar charts, pie charts, and histograms using collected data. Analyse the effectiveness of these diagrams for different types of data distributions.
2.	Represent raw data in frequency distribution tables for both ungrouped and grouped data by classifying data into intervals (for grouped data) and using tally bars. Calculate cumulative and relative frequencies for ungrouped and grouped data.
	Measure of Central Tendency
3.	Construct cumulative frequency curves and box-and-whisker diagrams for two different data sets. Calculate the median and quartiles for each data set. Compare and interpret the differences in central tendency and spread between the two data sets.
	Measure of Dispersion
4.	Calculate the coefficient of skewness using Karl Pearson's formula to identify the nature of skewness in a data set. Calculate the coefficient of kurtosis to classify a symmetrical distribution as platykurtic, mesokurtic, or leptokurtic.
	Regression and Correlation
5.	Apply the method of least squares to fit a regression line of Y on X and X on Y; interpret regression coefficient;
6.	Calculate the coefficient of correlation and determine the nature of correlation.

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Grade XII

S. No.	PRACTICAL ACTIVITY
	Random Variables
1.	Construct a probability distribution table for a given situation involving a discrete random variable and calculate mean and variance.
	Probability Distributions
2.	Apply the concept of the normal distribution to solve problems/ situation involving continuous data. Standardise data using z-scores, determine probabilities, and interpret areas under the normal curve.
	Sampling and Sampling Distributions
3.	Find the sampling distribution of the difference between two sample means drawn from independent populations. Compute the mean and standard deviation of the sampling distribution and verify key properties
	Estimation
4.	Estimate the confidence interval for the difference between the proportions of two populations using data from large independent samples. Apply appropriate statistical formulas to calculate the margin of error and interpret the results.
	Statistical Inference
5.	Apply hypothesis testing for a population mean in two scenarios: when the population standard deviation is known and when it is unknown. Formulate appropriate null and alternative hypotheses, calculate test statistics using the z-test and make decisions based on a given significance level also identify and interpret Type I or Type II errors.
6.	Calculate the degrees of freedom associated with the total sum of squares, the treatment sum of squares, and the error sum of squares in the context of analysis of variance (ANOVA).

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Scheme of Assessment

Grade XI

Table 1: Exam Specifications

Topic No.	Topics	Marks Distribution		Total Marks
		MCQs	CRQs	
1.	Introduction to Statistics	4	Total 2 Marks (1 CRQ)	6
2.	Presentation of Statistical Data	9	Total 4 Marks Choose any ONE from TWO (2 CRQs @ 4 Marks each)	13
3.	Measures of Central Tendency	10	Total 10 Marks Choose any TWO from THREE (3 CRQs @ 5 Marks each)	20
4.	Measures of Dispersion	10	Total 10 Marks Choose any TWO from THREE (3 CRQs @ 5 Marks each)	20
5.	Index Numbers	6	Total 3 Marks (1 CRQ)	9
6.	Time-Series	5	Total 3 Marks (1 CRQ)	8
7.	Regression and Correlation	6	Total 3 Marks (1 CRQ)	9
	Total	**50**	**35**	**85**
	Practical*			**15**
	Total Marks			**100**

Note: The cognitive distribution of marks for Statistics HSSC are as follows
Remember: 0 to 10 %
Understand: 40 to 60 %
Apply and beyond: 30 to 55 %

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Grade XII

Table 2: Exam Specifications

Topic No.	Topic	Marks Distribution		Total Marks
		MCQs	CRQs	
8.	Counting Techniques	7	Total 4 Marks Choose any ONE from TWO (2 CRQs @ 4 Marks each)	11
9.	Probability	7	Total 4 Marks Choose any ONE from TWO (2 CRQs @ 4 Marks each)	11
10.	Random Variables	10	Total 8 Marks Choose any TWO from THREE (3 CRQs @ 4 Marks each)	18
11.	Probability Distributions	12	Total 8 Marks Choose any TWO from THREE (3 CRQs @ 4 Marks each)	20
12.	Sampling and Sampling Distributions	5	Total 3 Marks (1 CRQ)	8
13.	Estimation	2	Total 3 Marks (1 CRQ)	5
14.	Statistical Inference	7	Total 5 Marks (1 CRQ)	12
	Total	**50**	**35**	**85**
	Practical*			**15**
	Total Marks			**100**

Note: The cognitive distribution of marks for Statistics HSSC are as follows
Remember: 0 to 10 %
Understand: 40 to 60 %
Apply and beyond: 30 to 55 %

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Examination Structure and Practical Requirements for Grades XI and XII

Theory:

- Multiple Choice Question (MCQ) requires candidates to choose one best/ correct answer from four options for each question. Each MCQ carries ONE mark.
- Constructed Response Question (CRQ) requires students to respond with a short text (few phrases/ sentences), calculations or diagrams.
- There will be two examinations, one at the end of grade XI and one at the end of grade XII.
- Table 1 and 2 contains the mark distribution for each topic.
- In each grade, the theory paper will be of 3 hours and will consist of two papers: paper I and paper II.
- Paper I theory will consist of 50 compulsory, multiple choice items. These questions will involve four responses options. The answer sheet for paper I will be provided separately.
- Paper II theory will carry 35 marks and consist of Constructed Response Questions (CRQs).
- The booklet for paper II will serve as an answer script.

Practical:

- In each grade, practical examination (Paper III) will be conducted separate from the theory paper and will consist of 15 marks.
- Practical examination (Paper III) will be based on the list of practical activities given in the examination syllabus. Schools may design their own practical manuals based on these activities for teaching and learning purposes.
- Practical journal/ portfolio should be developed by students and endorsed by a figure of authority, such as a teacher or principal, and submitted at the time of the practical examination (Paper III).

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